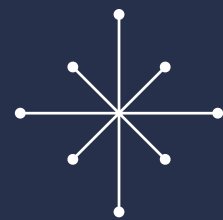


Presented by: Pavitra Pradeep
Muhammad Iqbal

BELOW THE WATERLINE

Climate
Adaptation in the
Netherlands



SDG 13 - Climate Action



WHAT IS THE PROBLEM?

- **10.7 million people in the Netherlands live in areas that could flood if primary flood defences fail, according to a 2026 Deltares calculation commissioned by Rijkswaterstaat.**

(Source: H2O Waternetwerk, “Niet 9 maar 10,7 miljoen Nederlanders lopen risico op mogelijke overstroming”, 20 March 2026. [H2O Waternetwerk](#))

- **Extreme rainfall and coastal pressure increase water-management challenges.**
- **Land subsidence adds pressure in vulnerable areas.**
- **Water infrastructure reduces risk, but adaptation is still needed.**



Maximum water depth in case of a dike breach
Source: Atlas Leefomgeving / Informatiehuis Water, 2025

How can low-lying Dutch communities continue adapting to increasing water-related risks while dealing with land subsidence?

TARGET AUDIENCE

What they already know

- Aware of climate change
- Familiar with Dutch water management
- Know the Netherlands has extensive water infrastructure

18–30 Young adults

*Living in urban areas of the
Netherlands*

Not primarily for:
Climate
scientists,
engineers, water-
management
professionals,
policymakers and
other specialists.

What they may not know

- Why adaptation is still necessary
- What land subsidence means
- How changing water risks affect Dutch communities

OUR SOLUTION

We created *Below the Waterline*, an AI-generated short film using ComfyUI that communicates the Netherlands' climate-adaptation challenges through visual storytelling.



“WE CAN’T STOP THE WATER. WE CAN ADAPT.”



The film presents three adaptation solutions:

- **Water-Resilient Neighborhoods:**

Rain gardens, water-retention areas, permeable spaces and raised walkways.

- **Floating Housing**

Floating homes as a climate-adaptive approach to living with changing water levels.



- **Climate-Resilient Cities**

Green rooftops, canals and water-retention basins integrated into urban design.

FROM RISK TO ADAPTATION

[LINK TO FILM](#)



9 shots • 30 seconds • AI-generated in ComfyUI

WHY DOES OUR SOLUTION FIT?

WHY DOES OUR SOLUTION FIT?

- **FAMILIAR:** Recognizable Dutch landscapes and water environments.
- **ACCESSIBLE:** Explains a complex issue in 30 seconds.
- **VISUAL:** Shows climate risks instead of technical explanations.
- **RELEVANT:** Connects adaptation to the communities where our audience lives.
- **SOLUTION-FOCUSED:** Shows resilient neighborhoods, floating housing and climate-resilient cities.



Why this audience?

Young adults(18-30) will live with the long-term effects of climate adaptation and the future development of Dutch communities and cities.

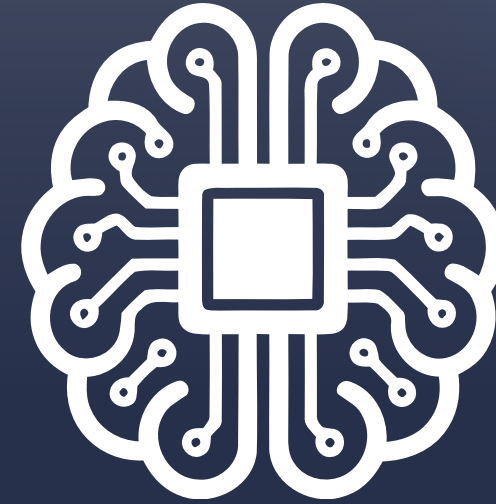
SDG-13 CLIMATE ACTION



How our project connects to SDG 13

- **Climate adaptation:** *Focuses on how the Netherlands can adapt to changing climate conditions.*
- **Extreme rainfall:** *Shows the pressure that heavy rainfall can place on low-lying areas and cities.*
- **Coastal water pressure:** *Highlights the continued challenge of living with water in a low-lying country.*
- **Land subsidence:** *Shows how sinking land adds to existing water-management challenges.*
- **Resilient communities:** *Presents adaptation through water-resilient neighborhoods and infrastructure.*
- **Future-focused solutions:** *Shows floating housing and climate-resilient urban design as examples of adaptation.*

ETHICAL CONSIDERATIONS



- **Photorealism:** AI visuals may be mistaken for real footage.
- **Fictional events:** Avoided unrealistic disaster and flooding scenes.
- **Real locations:** Dutch landscapes are shown as visual representations, not real events.
- **Privacy & likeness:** No identifiable people or likenesses were intentionally reproduced.
- **AI transparency:** Clearly disclosed the use of AI/ComfyUI.
- **Environmental impact:** Considered the energy and computational cost of AI generation.
- **Responsible communication:** Avoided sensationalizing climate risks.
- **Generation efficiency:** 11 generations → 9 final shots.

THANK YOU

for your time and attention